

Установка:

```
!pip install selenium webdriver-manager  
!apt-get update -qq  
!apt install -y chromium-browser chromium-chromedriver
```

Код тестов:

```
%%writefile test_schedule.py  
import pytest  
import time  
from selenium import webdriver  
from selenium.webdriver.common.by import By  
from selenium.webdriver.support.ui import Select, WebDriverWait  
from selenium.webdriver.support import expected_conditions as EC
```

```
@pytest.fixture  
def driver():  
    import google_colab_selenium as gs  
    driver = gs.Chrome()  
    driver.implicitly_wait(10)  
    yield driver  
    driver.quit()
```

```
@pytest.fixture  
def schedule_page(driver):  
    driver.get("https://api.nntu.ru/raspisanie")  
    WebDriverWait(driver, 15).until(  
        EC.presence_of_element_located((By.TAG_NAME, "select"))  
    )  
    time.sleep(3)  
  
    # Шаг 1: Форма обучения = Очная  
    selects = driver.find_elements(By.TAG_NAME, "select")  
    Select(selects[0]).select_by_visible_text("Очная")  
    time.sleep(5)  
  
    # Шаг 2: Группа = АСИ 23-1 (С ПРОБЕЛОМ!)  
    selects = driver.find_elements(By.TAG_NAME, "select")  
    Select(selects[1]).select_by_visible_text("АСИ 23-1")  
    time.sleep(3)  
  
    # Шаг 3: Вид расписания = Основное (если есть)  
    selects = driver.find_elements(By.TAG_NAME, "select")  
    try:
```

```
        Select(selects[2]).select_by_visible_text("Основное")
        time.sleep(3)
    except Exception:
        pass

    return driver
```

```
@pytest.mark.web
def test_page_loads(schedule_page):
    assert "Расписание" in schedule_page.title or "расписание" in
    schedule_page.page_source.lower()
```

```
@pytest.mark.web
def test_group_selected(schedule_page):
    selects = schedule_page.find_elements(By.TAG_NAME, "select")
    selected = Select(selects[1]).first_selected_option.text
    assert "АСИ 23-1" in selected
```

```
@pytest.mark.web
def test_schedule_table_present(schedule_page):
    tables = schedule_page.find_elements(By.TAG_NAME, "table")
    assert len(tables) > 0, "Таблица расписания не найдена"
```

```
@pytest.mark.parametrize("day, subject", [
    ("Вторник", "Надежность и отказоустойчивость информационных
систем"),
    ("Среда", "Анализ больших данных"),
    ("Четверг", "Технологии программирования"),
    ("Пятница", "Основы тестирования программного обеспечения"),
])
def test_subject_in_schedule(schedule_page, day, subject):
    page_text = schedule_page.page_source
    assert subject in page_text, f"Предмет '{subject}' не найден в расписании"
```

```
@pytest.mark.parametrize("teacher", [
    "Жидкова Наталья Валерьевна",
    "Рябов Антон Владимирович",
    "Емельянова Татьяна Владимировна",
    "Комаров Александр Олегович",
])
```

```

def test_teacher_in_schedule(schedule_page, teacher):
    page_text = schedule_page.page_source
    assert teacher in page_text, f"Преподаватель '{teacher}' не найден"

@pytest.mark.parametrize("auditorium", ["226", "324", "206", "320"])
def test_auditorium_in_schedule(schedule_page, auditorium):
    page_text = schedule_page.page_source
    assert auditorium in page_text, f"Аудитория '{auditorium}' не найдена"

@pytest.mark.web
def test_wrong_subject_fails(schedule_page):
    page_text = schedule_page.page_source
    assert "Квантовая физика" in page_text, "Ошибка: предмет найден, но его не должно быть"

@pytest.mark.skip(reason="Тест временно отключён")
def test_skipped_example():
    assert False

```

Вывод:

```

===== test session starts
=====

```

platform linux -- Python 3.13.15, pytest-8.4.2, pluggy-1.6.0 -- /usr/bin/python3

cachedir: .pytest_cache

rootdir: /content

configfile: pytest.ini

plugins: anyio-4.15.1, langsmith-0.12.5, typeguard-4.6.0

collected 17 items

test_schedule.py::test_page_loads PASSED [5%]

test_schedule.py::test_group_selected PASSED [11%]

test_schedule.py::test_schedule_table_present PASSED [17%]

[illegible]

```
test_schedule.py::test_auditorium_in_schedule[206] PASSED [ 82%]
test_schedule.py::test_auditorium_in_schedule[320] PASSED [ 88%]
test_schedule.py::test_wrong_subject_fails FAILED [ 94%]
test_schedule.py::test_skipped_example SKIPPED (Тест временно отключён)
[100%]
```

```
===== FAILURES
=====
```

```
_____ test_wrong_subject_fails
_____
```

```
test_schedule.py:96: in test_wrong_subject_fails
```

```
    assert "Квантовая физика" in page_text, "Ошибка: предмет найден, но его не
должно быть"
```

```
E  AssertionError: Ошибка: предмет найден, но его не должно быть
```

```
E  assert 'Квантовая физика' in '<html lang="en-US"><head><meta http-
equiv="origin-trial"
content="A7vZI3v+Gz7JfuRoIKNM4Aff6zaGuT7X0mf3wtoZTnKv6497cVMnhy
03KDqX7kBz/q/iidW7srW31oQbBt4VhgoAAACUeyJvcmlnaW4iOiJodHRwczov
L3d3dy5nb29nbGUuY29tOjQ0MyIsImZlYXR1cmUiOiJEaXNhYmxlVGhpcmR
QYXJ0eVN0b3JhZ2VQYXJ0aXRpb25pbmczIiwiaXhwaXJ5IjozU3OTgwOD
AwLCAJpc1N1YmRvbWFpbiI6dHJ1ZSwiaXNUaGlyZFBhcnR5Ijp0cnVlfQ==">\n
<meta charset="UTF-8">\n    <meta http-equiv="X-UA-Compatible"
content="IE=edge">\n    <meta http-equiv="content-type" content="text/html;
charset=utf-8">\n\t\t<meta name="yandex-verification"
content="d5312ad3909ee177">\n                <meta content="width=1024"
name="viewport">\n                <!-- <meta content="" name="keywords" />-->\n
<!-- <meta content="" name="description" />-->\n        <link
href="/frontend/web/img/favicon.ico" rel="shortcut icon" type="image/x-icon">\n
<!-- <script src="js/ie/reject.min.js" type="text/javascript"></script>-->\n
<!-- <link href="js/ie/reject.css" rel="stylesheet" />-->\n\t\t<!--
<style>\n\t\t\timgnewsmain{\n\t\t\t\t\tbackground-color: #ffffff!important;
\n\t\t\t\twidth: 100%!important;                /* Блок занимает всю
шири.../select2/select2.full.min.js" defer="defer"></script>\n<script
src="/frontend/web/js/custom.js" defer="defer"></script>\n<script
src="/frontend/web/js/revapi.js"
defer="defer"></script>\n<script>\n$(document).ready(function(){\n  var url =
location.href;\n  if(false){\n    bootstrap.alert('\n  });\n});\n\n</script>\n  <!--
```

```

— Yandex.Metrika counter ---->\n  <script type="text/javascript"> (function (d,
w, c) { (w[c] = w[c] || []).push(function() { try { w.yaCounter00000000 = new
Ya.Metrika({ id:00000000, clickmap:true, trackLinks:true,
accurateTrackBounce:true, webvisor:true }); } catch(e) { } }); var n =
d.getElementsByTagName("script")[0], s = d.createElement("script"), f = function
() { n.parentNode.insertBefore(s, n); }; s.type = "text/javascript"; s.async = true;
s.src = "https://mc.yandex.ru/metrika/watch.js"; if (w.opera == "[object Opera]") {
d.addEventListener("DOMContentLoaded", f, false); } else { f(); } })(document,
window, "yandex_metrika_callbacks"); </script> <noscript><div></div></noscript>\n  <!-- /Yandex.Metrika counter ---->\n
\n  \n</body></html>'

```

----- Captured stdout setup -----

<IPython.core.display.HTML object>

<IPython.core.display.Javascript object>

===== short test summary info
=====

FAILED test_schedule.py::test_wrong_subject_fails - AssertionError: Ошибка:
предмет найден, но его не должно быть

===== 1 failed, 15 passed, 1 skipped in 311.71s (0:05:11)
=====

Вывод с ошибкой:

===== test session starts
=====

platform linux -- Python 3.13.15, pytest-8.4.2, pluggy-1.6.0 -- /usr/bin/python3

cachedir: .pytest_cache

rootdir: /content

configfile: pytest.ini

plugins: anyio-4.15.1, langsmith-0.12.5, typeguard-4.6.0

collected 1 item

test_schedule.py::test_wrong_subject_fails FAILED

[100%]

===== FAILURES =====

test_wrong_subject_fails

test_schedule.py:96: in test_wrong_subject_fails

```
assert "Квантовая физика" in page_text, "Ошибка: предмет найден, но его не  
должно быть"
```

E AssertionError: Ошибка: предмет найден, но его не должно быть

```
E assert 'Квантовая физика' in '<html lang="en-US"><head><meta http-  
equiv="origin-trial"  
content="A7vZI3v+Gz7JfuRolKNM4Aff6zaGuT7X0mf3wtoZTnKv6497cVMnhy  
03KDqX7kBz/q/iidW7srW31oQbBt4VhgoAAACUeyJvcmlnaW4iOiJodHRwczoV  
L3d3dy5nb29nbGUuY29tOjQ0MyIsImZlYXR1cmUiOiJEaXNhYmxlVGhpcmR  
QYXJ0eVN0b3JhZ2VQYXJ0aXRpb25pbmczIiwiaXhwaXJ5IjoxNzU3OTgwOD  
AwLCJpc1N1YmRvbWFpbil6dHJ1ZSwiaXNUaGlyZF BhcnR5Ijp0cnVlfQ==">\n  
<meta charset="UTF-8">\n      <meta http-equiv="X-UA-Compatible"  
content="IE=edge">\n      <meta http-equiv="content-type" content="text/html;  
charset=utf-8">\n\t\t<meta name="yandex-verification"  
content="d5312ad3909ee177">\n          <meta content="width=1024"  
name="viewport">\n          <!-- <meta content="" name="keywords" />-->\n<!-- <meta content="" name="description" />-->\n      <link  
href="/frontend/web/img/favicon.ico" rel="shortcut icon" type="image/x-icon">\n<!-- <script src="js/ie/reject.min.js" type="text/javascript"></script>-->\n<!-- <link href="js/ie/reject.css" rel="stylesheet" />-->\n\t\t<!--  
<style>\n\t\t\timgnewsmain{\n\t\t\t\tbackground-color: #ffffff!important;\n\t\t\t\twidth: 100%!important;\n\t\t\t\t/* Блок занимает всю  
шири.../select2/select2.full.min.js" defer="defer"></script>\n<script  
src="/frontend/web/js/custom.js" defer="defer"></script>\n<script  
src="/frontend/web/js/revapi.js"  
defer="defer"></script>\n<script>\n$(document).ready(function(){\n  var url =  
location.href;\n  if(false){\n    bootbox.alert('\');\n  }\n});\n\n</script>\n  <!--  
— Yandex.Metrika counter —-->\n  <script type="text/javascript"> (function (d,  
w, c) { (w[c] = w[c] || []).push(function() { try { w.yaCounter00000000 = new  
Ya.Metrika({ id:00000000, clickmap:true, trackLinks:true,  
accurateTrackBounce:true, webvisor:true }); } catch(e) { } }); var n =  
d.getElementsByTagName("script")[0], s = d.createElement("script"), f = function  
() { n.parentNode.insertBefore(s, n); }; s.type = "text/javascript"; s.async = true;  
s.src = "https://mc.yandex.ru/metrika/watch.js"; if (w.opera == "[object Opera]") {  
d.addEventListener("DOMContentLoaded", f, false); } else { f(); } })(document,
```

```
window, "yandex_metrika_callbacks"); </script> <noscript><div></div></noscript>\n  <!-- /Yandex.Metrika counter ---->\n
\n  \n</body></html>
```

----- Captured stdout setup -----

<IPython.core.display.HTML object>

Hit:1 <http://security.ubuntu.com/ubuntu> noble-security InRelease

Hit:2 <https://cli.github.com/packages> stable InRelease

Hit:3 <https://cloud.r-project.org/bin/linux/ubuntu> noble-cran40/ InRelease

Hit:4 <https://dl.google.com/linux/chrome-stable/deb> stable InRelease

Hit:5 <http://archive.ubuntu.com/ubuntu> noble InRelease

Hit:6 <https://r2u.stat.illinois.edu/ubuntu> noble InRelease

Hit:7 <http://archive.ubuntu.com/ubuntu> noble-updates InRelease

Hit:8 <http://archive.ubuntu.com/ubuntu> noble-backports InRelease

Hit:9 <https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu> noble InRelease

Reading package lists...

<IPython.core.display.Javascript object>

<IPython.core.display.HTML object>

Reading package lists...

Building dependency tree...

Reading state information...

google-chrome-stable is already the newest version (154.0.8037.97-1).

0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.

<IPython.core.display.Javascript object>

<IPython.core.display.HTML object>

<IPython.core.display.Javascript object>

----- Captured stderr setup -----

W: Skipping acquire of configured file 'main/source/Sources' as repository
'https://r2u.stat.illinois.edu/ubuntu noble InRelease' does not seem to provide it
(sources.list entry misspelt?)

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload	Upload	Total	Spent	Left
100	135M	100	135M	0	0	254M	0
						--:--:--	--:--:--
						--:--:--	--:--:--
						--:--:--	254M

===== short test summary info
=====

FAILED test_schedule.py::test_wrong_subject_fails - AssertionError: Ошибка:
предмет найден, но его не должно быть

===== 1 failed in 25.83s
=====

Пояснение применённых техник

- Базовые тесты (3 шт.): test_page_loads, test_group_selected, test_schedule_table_present
- Параметризованные тесты (3 шт. × несколько значений): test_subject_in_schedule, test_teacher_in_schedule, test_auditorium_in_schedule
- Фикстуры: driver (запуск браузера), schedule_page (открытие расписания с выбранной группой)
- Метки: @pytest.mark.web
- Пропускаемый тест: test_skipped_example (SKIPPED)
- Тест с неправильным ожиданием: test_wrong_subject_fails — падает, демонстрируя корректную работу механизма проверки